

	Therefore, $\dots = 0.5$
10	Ans : B At a height of 1.0×10^6 m, distance from the centre of the Earth, $r = 1.0 \times 10^6 + 6.37 \times 10^6 = 7.37 \times 10^6$ m From Conservation of Energy, total energy at r = total energy at infinity = 0 $-\frac{GMm}{r} + \frac{1}{2}mv^2 = 0$ $v = \sqrt{\frac{2GM}{r}} = \sqrt{\frac{2 \times 6.67 \times 10^{-11} \times 5.96 \times 10^{24}}{7.37 \times 10^6}} = 1.04 \times 10^4 \text{ m s}^{-1}$

3

11

Ans : B